Team Name at SemEval-2027 Task 1: Multi-evidence re-ranking with a frozen first stage

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Abstract

RETECO Track 1a asks for documents that answer a *complete* question about a historical period: long queries whose temporal and exclusion conditions sit in the body rather than the title, and judgements that mark **several** documents per question (3.44 on average, over **2.34 annotated steps**). We describe a two-stage ranker - a zero-shot dense retriever plus a windowed cross-encoder fine-tuned on the 1,071 training queries that carry at least one positive among their dense top-100 candidates - that scores 0.3581 domain-macro nDCG@10 on the 13-domain dev set, against 0.2943 for the dense stage and 0.1147 for the organisers' BM25 baseline. Against the same pipeline with the cross-encoder left at zero-shot (0.3249), the fine-tuned reranker contributes +0.0332 (domain-clustered 95% CI [+0.0192, +0.0472]) at a fusion form fixed before fine-tuning, and reproduces at +0.0326 / +0.0279 under two further seeds. The remaining headroom is only partly recall: at the 100 candidates the reranker sees, 80.7% of annotated steps have at least one relevant document (96.3% within the dense top-1000), yet the fused top ten covers only 59.5% of the annotated steps and the cross-encoder's own top ten 51.4%. Ten temporal probes, one strictly pre-registered test (+0.0039 against a +0.02 bar) and a fixed year-consistency reordering that *lowers* nDCG@10 in 7 of 8 domains do not close it. Supervising every annotated step to have a positive ranked high costs 0.014 domain-macro and *lowers* coverage, 59.5% to 55.6% fused and 51.4% to 48.1% for the cross-encoder alone. A constraint-faithful 64-word query and the restoration of the 140 training queries with an empty candidate pool each cost 0.0138 - but only 0.0026 against a same-code retrain of the unchanged objective, which itself costs 0.0111, so on dev they are indistinguishable from it. Every intervention in the six-step ladder is neutral or negative, and the frozen system is kept.

1 Introduction

Track 1a gives 13 independent corpora, long queries, binary qrels and no timestamps. Our entry has two parts: a strong, reproducible zero-shot-to-supervised pipeline, and a diagnosis of why the obvious temporal fixes do not work.

Contributions: (i) a baseline ladder over all 13 domains under the official domain-macro statistic, with paired bootstraps over queries and over domains; (ii) a supervised cross-encoder that adds +0.0332 without tuning the fusion weight on dev; (iii) a multi-evidence analysis showing that the system's failure mode is "one easy document" inside a candidate pool that is itself incomplete; (iv) negative evidence on explicit temporal modelling: one pre-registered test and a fixed year-consistency rule that fails; (v) a decomposition of why the cross-encoder with the best standalone score is not the one that fuses best - anchoring, attenuation and score calibration - and what that implies for using a single-stage metric to select checkpoints.

Relation to prior work. The reranker is the standard retrieve-then-rerank cross-encoder [4], initialised from M3-Embedding [10]. Our multi-evidence objective is a sibling of listwise and LambdaRank training [6, 7], with one difference: instead of ordering one list, it asks that every official step of a question own a document in the top ten. Multi-stage fusion has its own line - list-aware reranking of the first stage's list [12] and joint selection of documents that cover the answer set [13] - but both learn the second stage jointly with the list it consumes (HLATR adds a list-aware module over retrieved and reranked lists; JPR reranks the set autoregressively), whereas we freeze the first stage and ask what one second stage can add at a fixed fusion weight. The contribution is the measured behaviour of that frozen system, not a new fusion architecture. Temporal retrieval is the benchmark's own subject [1]; our contribution there is negative

evidence about what does not transfer from the two-domain pilot to the full suite.

2 Task, metric, data

Metric. Official aggregation is domain-macro nDCG@10: per-query nDCG@10, averaged within domain, then over the 13 domains, so a 3-query domain weighs as much as a 240-query one, and queries missing from a run score zero. dev = 519 queries. 26.6% of dev questions receive exactly zero from the frozen system, and 37.7% of those have no relevant document in the top 100 at all. Those zeros dominate the statistic: had the 138 zero questions merely reached half their domain's average, domain-macro would be 0.4283 instead of 0.3581 (full average: 0.4985, `results/ZERO_HEADROOM.json`).

Data. 1,167,159 documents, 1,211 train queries, binary qrels (so nDCG@10 is discounted precision at 10). Per-domain counts in `results/corpus_manifest.json`. Queries carry a title and a body separated by <p>; the body holds the temporal range, exclusions and comparisons (Section 5.3).

3 System

Dense first stage. `mxbai-embed-large-v1`, full query with the model's retrieval instruction, mean pooling, cosine similarity, fp16, `top-100` candidates per query. Dev: 0.2943 (0.2956 with CLS pooling: a wash).

Windowed cross-encoder. `bge-reranker-v2-m3` (568M). Documents are split into 250-token windows with stride 192, each window is scored against the query, and the document score is the maximum over its windows, i.e. a document is as relevant as its best passage; query capped at 256 tokens. Pooling was fixed a priori and is not ablated (Section 7). Zero-shot at fusion 1:1: 0.3249.

Fusion. Per-query z-normalisation of the dense and cross-encoder lists, combined at 1:1. The weight was fixed on the training split before any dev inspection; Section 5.1 shows the choice is not what produces the gain.

Supervised fine-tuning. Listwise softmax over a 20-document group sampled from each training query's 100 candidates - 21,420 query-document pairs in 1,071 groups, the log-sum-exp running over the group's positives - 8 of 24 layers frozen, lr 1.5e-5, seed 17, 2 epochs; epoch 1 was kept, by a 0.3029-against- 0.3012 internal-validation margin. Checkpoint selection used the cross-encoder's own ordering on 45 held-out training queries, not the fused system; that run predates domain-macro logging and selected on the plain mean, and on every later arm the two criteria pick the same epoch. Frozen dev: **0.3581**.

system	dev domain-macro nDCG@10	Δ vs zero-shot z1:1	domain-clustered 95% CI
official BM25	0.1147	—	—
dense (mxbai, top-100)	0.2943	—	—
dense + zero-shot CE, z 1:1	0.3249	—	—
+ SFT CE, 1 epoch	0.3336	+0.0087	[+0.0037, +0.0142]
+ SFT CE, 2 epochs (frozen)	**0.3581**	**+0.0332**	**[+0.0192, +0.0472]**
same recipe, seed 23	0.3575	+0.0326	[+0.0184, +0.0470]
same recipe, seed 29 (two independent reruns)	0.3528 / 0.3527	+0.0279	[+0.0160, +0.0405]

Multi-evidence (step) supervision. Track 1a gives one relevance judgement per query, but the organisers' qrels also carry *step* annotations: a question that needs several pieces of evidence is decomposed into steps, and each step lists the documents that satisfy it. We use them only in training; at test time the ranker still sees the full question and one document. With `P_j` the step-`j` positives inside a query's candidate list, the ordinary listwise term `-log sum_{d in P} p(d)` is dominated by the single easiest positive, so we add

` L_step = -(1/m) * sum_j log sum_{d in P_j} p(d), L = L_list + lambda * L_step `

which forces *every* step to have a positive ranked high instead of letting one easy document carry the query. Steps whose positives are absent from the candidate list are skipped (a step is covered for 91.2% of step instances under our candidate construction); `m` counts the covered steps. `lambda` is not tuned on dev: two values (0.3, 1.0) are trained on the same groups, and both raise the train-internal split above the baseline (0.2882 and 0.2913 against 0.2865), yet both are rejected on dev (S5.6).

4 Results

The gain holds when the three smallest domains are dropped (0.3267 against 0.2985 on domains with n >= 19 queries) and in 11 of 13 domains; the two exceptions, bitcoin and quant, are flat (-0.0023 and -0.0009). Seed spread over three seeds is 0.0029, an order of magnitude below the effect, while a fourth run of the same recipe through the current code path scores 0.3470: the retraining band is 0.347-0.358 and the frozen artifact sits at its top (S5.5). Rank-fusion forms were screened too and are reported in S5.1, where the RRF list that out-of-fold selection prefers is worse than the z 1:1 fusion used here.

5 Analysis

5.1 Is the gain an artefact of the fusion weight?

No. A sweep over nine weights (0.5 to 2 on the cross-encoder, z-normalised) is positive everywhere (+0.0089 to +0.0346; the w=1 row reproduces the headline within bootstrap noise), and refitting the weight out-of-fold on 45 held-out training queries selects a dense-heavy RRF 1:4 that is *worse* than z 1:1, so the gain is not bought by tuning on dev.

5.2 The bottleneck is multi-evidence coverage over an incomplete pool

We measured, per system and without extra annotation, how much of the official step structure reaches the top ten (`scripts/dev_multievidence.py`):

Supervision improves every column but leaves most of the structure unused. At depth 1000 seven of ten steps are retrievable and still four of ten never reach the top ten; at the depth the reranker sees (100 candidates per query) only 80.7% of annotated steps are reachable and 88.3% of questions keep a positive, so a fifth of the step structure is lost before the cross-encoder runs. Truncating the fused list from 1000 to 100 rows leaves its top ten unchanged (59.45%, `results/MULTIEVIDENCE_sft_e2bc_sub100.json`), so depth moves reachability only; the 51.4% is the cross-encoder's own ordering over those 100 candidates. The retrieval floor is small but visible: in 10 of 519 dev questions (1.9%) no relevant document appears anywhere in the dense top-1000, and 52 of the 138 questions the frozen system scores at zero (38%) have none in the top 100, where no reranker can lift them. Coverage tracks nDCG@10 (Spearman 0.69; all-covered questions average 0.595 against 0.187, `results/STEP_COVERAGE_CORR.json`), though partly mechanically, so we read it as a check rather than a result.

The gain does not concentrate on the questions that need several evidence documents: on the 151 two-step dev questions it is +0.0200 (CI [-0.0033, +0.0463]) against +0.0711 on single-step ones (Appendix A.4).

5.3 Query truncation drops the conditions

In 1,730 train+dev queries, 44.8% (46.4% of dev: 241 of 519) carry at least one condition (year 14.3%, temporal relation 19.0%, exclusion 16.3%, comparison 14.0%) that appears **only in the body**, so it never reaches the cross-encoder, which is trained on the title alone; the dense stage reads the full question. That truncation is not, by itself, what the ranker is missing: on the 241 condition-lost queries the frozen system scores **0.3656 against 0.3518** on the rest, and its gain over zero-shot CE is about twice as large there (+0.0481 vs +0.0246), although the dense stage already reads the full question and only the reranker does not. A same-budget body-visible composer therefore has to *earn* the claim, so we pre-register that its gain must concentrate on the lost subset. Two 64-word composers were built - a cue-sentence selector (title plus only the body sentences carrying a condition) and the plain full question (title plus the first 64 body words). On the 45-query internal split no representation wins outside noise: title 0.2827, full 0.2869, selector 0.2744, fixed body 0.2703, frequency summary 0.2614 (`results/QVARIANT_WINNER.json`); the ~+-0.03 resolution of that split (S7) makes dev the decisive test, and the selector - the more faithful of the two - was the worst, so dev carries the *generous* arm. Trained end to end on the 64-word full question it scores **0.3443** against 0.3581 (Delta -0.0138, query-clustered CI [-0.0272, -0.0005], domain-clustered [-0.0338, +0.0052]) and loses on both subsets: 0.3566 vs 0.3656 on the 241 condition-lost queries and 0.3296 vs 0.3518 on the other 278. The loss is thus *smaller* where the arm was designed to help (-0.0090 vs -0.0222), consistent with body conditions not being the missing signal, but no subset gains, so the pre-registered bar is not met (`results/CONSTRAINT_SUBSET_cx_qv.json`).

5.4 Temporal modelling did not help

Ten probes - year/anchor features, a semantic time-consistency score, oracle and LLM step decomposition, HyDE, temporal hard negatives, a time-aware constraint rerank, a train-fitted gate, a fixed year-consistency reordering, and a coverage diagnostic. Exactly one was strictly pre-registered: it gained +0.0039 ([-0.0023, +0.0103], domain [+0.0006, +0.0142], P=0.893) against a +0.02 bar. The year rule fails cleanly: ordering the frozen candidates by year consistency lowers nDCG@10 in 7 of 8 anchored domains (macro 0.2980 to 0.2498). That is a fixed rule losing, not an upper bound - the reordered list is not the best possible one, and the rule itself is what a temporal ranker would have to beat. In this corpus a document mentioning another period is often still the right answer.

system	nDCG@10	relevant docs in top-10	questions with >=2	step coverage of top-10	steps retrievable @1000
dense	0.2943	1.03	31.8%	51.3%	96.3%
+ zero-shot CE	0.3249	1.11	35.6%	54.9%	96.3%
+ SFT CE (frozen)	0.3581	1.21	38.3%	59.5%	96.3%

5.5 The rest of the ladder, in brief

Three further steps are reported in full in Appendix A. **Loss forms**: averaging the log-loss over all positives (0.2809), an nDCG-weighted permutation loss (0.2764) and domain-sqrt sampling (0.2912) all sit within the 0.010 tie band of the current objective (0.2865) on the 45-query split, and dev separates none of them from the unchanged objective re-run through the current code path (0.3470, 0.0111 below the frozen artifact). **Step supervision**: requiring every annotated step to have a positive ranked high costs 0.0139 / 0.0143 and *lowers* the coverage it targets (59.5% to 55.6% / 56.0%), even on the two-step questions (0.4058 to 0.3910 / 0.3889). **Corrections**: the 64-word full question and the restoration of the 140 empty-pool queries each cost 0.0138, but only 0.0026 against the same-code retrain, and every interval includes zero (`results/ATTRIBUTION_LX-POSSUM.json`). With S5.3 and S5.4 the message is consistent: every correction is neutral or negative here, and the ladder's five retrained steps never beat 0.3581.

5.6 A better cross-encoder is not a better system

The checkpoint screen scores the cross-encoder alone; the system is the dense+CE fusion, and the two rank the arms differently. `cx_qv` is the stronger cross-encoder by 0.0397 (0.3153 against 0.2756) and 0.0027 *worse* fused at the frozen weight; across the six arms scored, the best CE-only arm is third fused and the second-best is first (Spearman 0.66).

Three mechanisms explain it (519 dev queries, frozen dense top-1000, per-query z-normalised 1:1 fusion). *(i) Anchoring*: the fused top ten is 74% dense documents and only 55-60% of it is also in the CE top ten, so a changed ordering must displace dense documents to matter. *(ii) Attenuation*: a CE gain transfers at slope 0.30 - +0.1985 on the 185 queries the CE improves becomes +0.0458 fused, and -0.1398 on the 113 it hurts becomes -0.0482. *(iii) Calibration*: on the 221 queries where the CE's own nDCG@10 does not move (|deltal <= 0.01) the fused score still falls 0.0116, and a transplant isolates the two - `cx_qv`'s ordering with `lx_possum`'s score profile scores 0.3482 (+0.0012) while the reverse falls to 0.3403 (-0.0067).

Re-weighting does not fix it. The optimum is model-specific (`cx_qv` wants w = 2, worth +0.0103 [+0.0012, +0.0201] over w = 1, a quarter of the CE gain), but the weight must be chosen somewhere: chosen on the 45 held-out queries `cx_qv` reaches 0.3509, still -0.0072 against the frozen system, and the same procedure costs `lx_possum` 0.0058. A calibration-free RRF (k = 60) recovers +0.0069 and trails by 0.0119. On those queries the model-level inversion reproduces with the candidate list fixed (`cx_qv` 0.2893 alone against 0.2462, and 0.3197 against 0.3325 fused); within a model both criteria prefer epoch 1, and that preference is itself noise (epoch 1 on dev: 0.3428 against 0.3470, CI [-0.0114, +0.0022], 397 of 519 tied). A single-stage metric is thus a good ranker of cross-encoders and a poor ranker of the system they feed.

7 Limitations

Domain-macro lets 3-query domains weigh as much as 240-query ones, so a single-domain regression can move the headline; the temporal findings are about *textual* time only, not about systems that have timestamps; and the supervised results are single-corpus and single-family (one dense model, one cross-encoder). The window/pooling schedule (250-token windows, stride 192, max over windows, 256-token query cap) was fixed a priori and is not ablated; the dense encoder's pooling was ablated late and is a wash (CLS 0.2956 against 0.2943 for mean); the 1:1 fusion weight and the two-epoch schedule were picked on the training-internal split, so the frozen number carries that selection. The step-supervision results are reported at one seed; a second seed and distillation of a larger reranker [9] were not affordable within the GPU budget.

Selection is noisy by construction: the 45-query internal split improves the sum-positive arm by +0.0135 with a domain-clustered interval of [-0.0317, +0.0571], i.e. it cannot resolve differences below about 0.04. Every arm here is therefore screened internally and judged on dev with a paired domain bootstrap (half-width 0.007-0.014); dev gains below +0.010 are treated as unresolved. The

failure	query	gold	dense rank	nDCG@10
pool miss	travel/201562_1094	2	656, not retrieved	0.0000
ordering	history/70583_271	1	24 (CE 14, fused 17)	0.0000
step coverage	history/59309_621	2	1, 10 (CE 1, 72)	0.6131

screen did not promote every loser (S5.5) - what it cannot do is rank these arms: the four it scored lie within 0.006 of each other, dev puts them 0.0026-0.0031 apart from the same-code baseline, and every interval includes zero. Retraining is not free: fresh pools and seeds give 0.3575 / 0.3528 / 0.3527 and the current code path 0.3470, so 0.3581 belongs to the artifact, not to the recipe.

8 Error analysis

A quarter of dev questions (138/519, 26.6%) score exactly zero. The split separates cleanly by candidate-pool recall: **52 (37.7%) have no relevant document in the dense top-100** - no top-100 reranker can score them - and 86 (62.3%) have the gold in the pool and lose it to ordering. Zeros cluster where the corpus is thin (history 46, politics 14, bitcoin 13). Three worked examples, one per mechanism (`results/DEV_ERROR_ANALYSIS.md`):

The pool miss is a PNR booking-limit question whose evidence sits outside the 100 candidates the reranker sees (dense rank 656); a deeper second stage could reach that document, but nothing downstream of the reranker can. The ordering case ("How do we know when Nero was born?") has its gold at dense rank 24, moves to 14 for the cross-encoder and 17 fused, and scores zero. The step case ("How many people watched the moon landings after Apollo 11?") keeps the Apollo 11 figures at rank 1 but pushes the comparison document from dense rank 10 to 72 for the title-only cross-encoder: one of two evidence documents survives, nDCG@10 stays at 0.61, and step coverage fails. That shape is common, not anomalous: only 38.3% of questions place two or more relevant documents in the top ten (S5.2).

9 Conclusion

A zero-shot dense stage plus a cross-encoder fine-tuned on the 1,071 questions with an in-pool positive gives +0.0332 domain-macro nDCG@10 over the same pipeline with the cross-encoder left at zero-shot (0.3249), on all 13 dev domains, stable across seeds (0.347-0.358). Two failures remain: an incomplete pool (1.9% of questions have no gold in the dense top-1,000) and multi-evidence coverage (59.5% of steps in the fused top ten). Every explicit temporal intervention - ten probes, one pre-registered test, step supervision, query reformulation, loss variants, backfilling, rank fusion - is neutral or negative. Most transferable: a cross-encoder 0.04 better on its own metric can be worse fused, because fusion anchors on the first stage and re-normalises the second's scores.

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objective / sampler	best epoch	val domain-macro nDCG@10
current (log-sum-exp over positives)	2	0.2865
mean over positives	2	0.2809
nDCG-weighted permutations	2	0.2764
current + sqrt(domain) sampling	2	0.2912
zero-shot reference	-	0.2730

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Appendix A. The six-step ladder in full

A.1 The loss form is not the bottleneck

Three alternatives suggested by the coverage gap - averaging the log-loss over all positives, an nDCG-weighted permutation loss (LambdaLoss-style, cut-off 10), and domain-balanced (sqrt) sampling - were trained on the same 942 groups with the same hyper-parameters and scored on the same 45-query split:

Selecting on domain-macro rather than the plain mean picks the same epoch in all four arms, so the official aggregation does not move the checkpoint; the split cannot resolve the gaps (+-0.03, S7), so only sqrt sampling's +0.0047 ([-0.0164, +0.0285]) is even measurable, and the pre-registered 0.010 tie band keeps the current objective. Sqrt sampling's one visible effect is raising the share of queries with a relevant document in the top ten from 93.3% to 95.6% without moving nDCG: it makes the easy decision easier.

Dev separates two effects this screen conflates. The incumbent objective retrained through the current code path (same 942 groups, two epochs, each run taking its own val-preferred epoch) scores **0.3470**, 0.0111 below the frozen artifact (domain-clustered CI [-0.0265, +0.0057], 132 better / 167 worse). That gap is the price of re-running the recipe, not of a new objective, and it is the baseline against which the four arms should be read. They stay unmeasured on dev - the screen separates them by at most 0.015, below the dev half-width of 0.007-0.014 - so what dev supports is narrower than "the replacements lose": the screen ranks arms at 0.2764-0.2912, cannot resolve 0.003, and the one arm it put on dev, the incumbent itself mid-screen at 0.2865, reproduces 0.0111 below the frozen number.

A.2 Forcing every evidence step into the top ten does not pay

If the fused top ten covers only 59.5% of annotated steps (51.4% for the cross-encoder alone), the mechanical remedy is to require every step to have a positive ranked high. We trained the step-augmented objective of S3 at two weights and scored the systems on dev with the frozen fusion:

The coverage column counts the fused list, whose top ten does not change when the list is truncated from 1000 to 100 rows (59.45%); the cross-encoder's own ordering over the same candidates covers 51.39% before the term and 48.14% / 45.78% with it, so the degradation is a property of the ordering, not of depth.

The term does not merely fail to pay: it *reduces* the quantity it optimises. Both arms also lose on the two-step questions they target (0.4058 to 0.3910 and 0.3889, CI including zero), and the loss is spread over all partitions. Two readings fit: a step's positives are mostly documents the ordinary list-wise term already covers (3.44 relevant documents over 2.34 steps, so a step rarely adds a *different* document), and the log-sum-exp inside a step is dominated by the step whose positive is hardest to retrieve, so the gradient chases documents that are absent or buried. The annotations are not at fault: every one of the 2,873 step-level positives is also a query-level positive. The instructive part is the selection: on the internal split lambda = 1.0 was the *best* of six arms trained that day (0.2913, above 0.2865), and dev puts it last - which is why S3 fixes the fusion form in advance and S7 refuses to select on 45 questions.

A.3 Two further corrections that also lose

The frozen system feeds the reranker a title-only query and excludes the 140 training queries whose dense top-100 contains no positive. Both were corrected with the frozen recipe and scored on dev with the frozen fusion:

The two cost exactly the same 0.0138, which suggests a flat basin in both directions rather than two intrinsically bad ideas. For the query arm the loss is smaller where the body conditions live (S5.3), so the extra text carries signal but not enough to pay for the harm on the other 278 queries. For the backfill arm the 294 restored positives are, by construction, documents the first stage never retrieves, so training on them spends capacity on a decision never made at inference, where the candidate list is the dense top-100. Most of the 0.0138 is not the correction anyway: the unchanged objective re-run through the current code path costs 0.0111 (Appendix A), and against that baseline the corrections cost 0.0026

system	dev domain-macro	Delta vs frozen	domain-clustered 95% CI	steps in top 10	questions with 2+ relevant in top 10
frozen (no step term)	0.3581	-	-	59.45%	38.31%
L_step, lambda = 0.3	0.3442	-0.0139	[-0.0348, +0.0044]	55.58%	37.95%
L_step, lambda = 1.0	0.3438	-0.0143	[-0.0351, +0.0037]	56.00%	37.88%

system	dev domain-macro	Delta vs frozen	query-clustered 95% CI	domain-clustered 95% CI
frozen	0.3581	-	-	-
full-question query, 64-word cap	0.3443	-0.0138	[-0.0272, -0.0005]	[-0.0338, +0.0052]
+ 140 backfilled queries, 294 positives	0.3443	-0.0138	[-0.0288, +0.0010]	[-0.0311, +0.0012]

each (domain-clustered [-0.0187, +0.0100] and [-0.0162, +0.0101]); the step arms behave the same way (-0.0027 [-0.0154, +0.0082] and -0.0031 [-0.0153, +0.0072], `results/ATTRIBUTION_LXPOSSUM.json`). Dev therefore supports a deployment decision - five of the ladder's six steps retrained, none beating 0.3581, and the sixth (dense pooling) a wash at +0.0014 - plus a negative result against a same-code baseline that the 45-query screen cannot rank at all.

A.4 Where the gain sits: not on the multi-step questions

Partitioning dev by the number of official evidence steps a question needs (`results/STEP_SPLIT.md`), the frozen system's gain over zero-shot CE is largest on single-step questions and weakest on two-step ones:

The objective buys a better *single-document* decision almost everywhere but does not teach evidence assembly: on the 151 two-step questions the gain is within noise. That is the gap the step term targets, and it fixes the bar those arms have to clear.

partition	queries	zero-shot CE	frozen SFT	delta	95% CI (domain-clustered)
1 step	75	0.3185	0.3896	+0.0711	[+0.0244, +0.1056]
2 steps	151	0.3857	0.4058	+0.0200	[-0.0033, +0.0463]
3+ steps	258	0.3196	0.3511	+0.0315	[+0.0084, +0.0571]
none	35	0.1579	0.1705	+0.0126	[+0.0030, +0.0524]